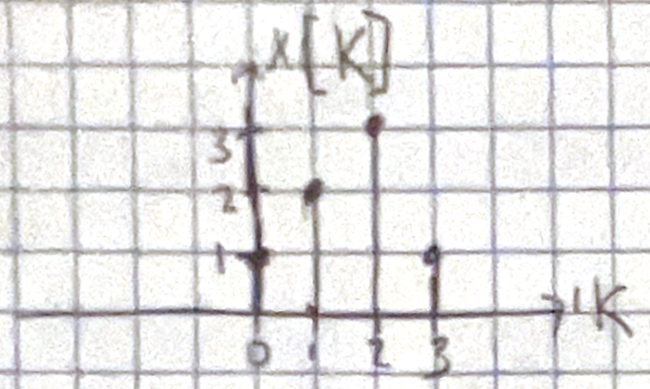
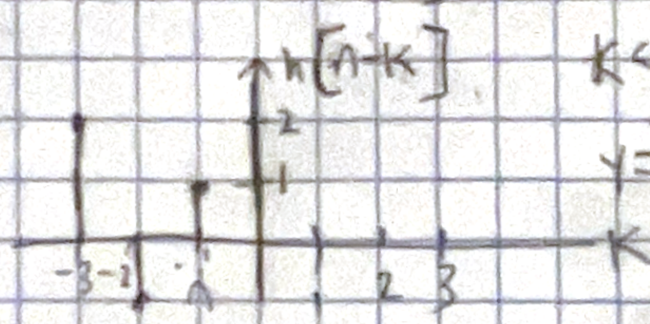
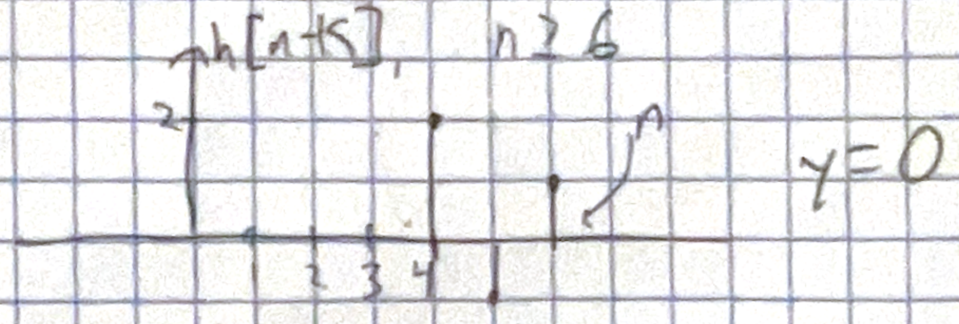




$k<0$

$y=0$

$$y[n]=\sum_{k=-\infty}^{\infty} x[k]h[n-k]$$



$y=0$

$h[n-k]$, $n=0 \Rightarrow h[-k]=\{2,-1,1\}$, $k=-2 \dots 0$

$y[0]=1(1)=1$

$h[n-k]$, $n=1 \Rightarrow h[1-k]=\{2,-1,1\}$, $k=-1 \dots 1$

$y[1]=1\cdot(-1)+2\cdot(1)=-1+2=1$

$h[n-k]$, $n=2 \Rightarrow h[2-k]=\{2,-1,1\}$, $k=0 \dots 2$

$y[2]=1(2)+2(-1)+3(1)=2-2+3=3$

$h[n-k]$, $n=3 \Rightarrow h[3-k]=\{2,-1,1\}$, $k=1 \dots 3$

$y[3]=1(0)+2(2)+3(-1)+1(1)=4-3+1=2$

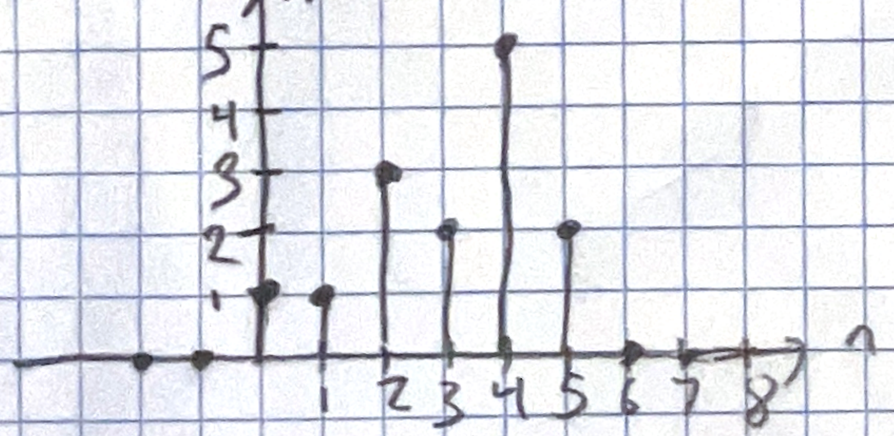
$h[n-k]$, $n=4 \Rightarrow h[4-k]=\{2,-1,1\}$, $k=2 \dots 4$

$y[4]=1(0)+2(0)+3(2)+1(-1)=6-1=5$

$h[n-k]$, $n=5 \Rightarrow h[5-k]=\{2,-1,1\}$, $k=3 \dots 5$

$y[5]=1(0)+2(0)+3(0)+1(2)=2$

$y[n]=\{1,1,3,2,5,2\}$ for $n=0,\dots 5$, $y[n]=0$ otherwise



$N_x=4$, $N_h=3$

$N_y=6$

$N_x+N_h-1=4+3-1=6=N_y$

$\rightarrow$ length of y is as expected (6)